Case Study Plan: Managing a Software Database for the Web Development Lab at Florida Atlantic University (FAU)

Sean Heller

Introduction

Imagine walking into the Web Development Lab at Florida Atlantic University (FAU) minutes before a major project deadline, only to discover that the software version on your assigned computer does not match the one you used in class. This has become a common frustration for students in the Information Science – Web Development concentration, where inconsistent software installations and outdated tools frequently disrupt coursework and capstone projects. These recurring issues highlight the need for a structured, reliable system to track and manage the lab's software environment.

Project management provides a systematic approach for addressing such challenges. In the Project Management Handbook, (Baars, 2006) outlines a six-phase model (initiation, definition, design, development, implementation, and follow-up) that guides projects from concept to completion in a controlled and organized manner. In this case study, I apply this model to a small but meaningful project: developing a centralized software-tracking database for FAU's Web Development Lab. By following this structured framework, I lead a team of senior Information Science students in designing and implementing a solution that improves organization, supports instruction, and enhances the student learning experience.

Initiation Phase

The Project Management Handbook explains that the initiation phase is where the idea for a project is explored and its feasibility is evaluated (Baars, 2006). In the Web Development Lab, the need for a software-tracking system became very clear to me after repeated issues with inconsistent software versions, missing tools, and student project failures caused by mismatched configurations. Faculty members reported that students often spent more time troubleshooting lab computers than working on their assignments.

To confirm the project's necessity, I met with the faculty advisor for the Web Development concentration, the lab's IT liaison, and instructors who regularly use the lab. These discussions helped me identify the core objective: creating a database that tracks all software installed on the lab's computers, including versions, installation dates, license types, and course dependencies. Following the guidelines of the handbook, I prepared a project proposal outlining the problem, expected benefits, estimated timeline, and required resources. In result of this phase, I was able to get my project approved allowing my team to proceed.

Definition Phase

The definition phase focuses on clarifying the project's scope, requirements, and risks. According to the six-phase model presented in the Project Management Handbook (Baars, 2006), clearly defining requirements early prevents confusion and delays later in the project. Because this project is small, I limited the scope to 20 computers in the Web Development Lab and the software used in upper-level Information Science courses.

I organized structured meetings with my student team to determine the exact data fields needed in the database. We identified fields such as software name, version, developer, license type, installation date, machine ID, and associated courses. We also agreed that a simple web-based interface would be necessary so faculty and IT staff could easily view and update records.

I created a Work Breakdown Structure (WBS) to divide the project into manageable tasks, including requirements gathering, database design, interface development, testing, and documentation. I also conducted a risk analysis to identify potential challenges, such as incomplete software lists, difficulty accessing lab machines, and inconsistent installation histories. In result of this phase, I was able to deliver a detail document of requirements and got a clearly defined slope statement approved by the faculty advisor and lab IT liaison.

Design Phase

In the design phase, the project's requirements are translated into a structured technical solution. As described in the Project Management Handbook (Baars, 2006), this stage ensures that all major decisions are made before development begins. To accomplish this, I collaborated with my team to design the database architecture, user interface, and workflow for updating software records.

The database developer on my team created an entity-relationship diagram showing how software records relate to machines, license types, and courses. Our UI/UX designer produced mockups for a simple dashboard that allows users to search for software, filter by machine, and view version histories. I also ensured that security considerations were addressed by restricting modification permissions to authorized faculty and IT staff.

Before moving to development, I presented the design documents to the faculty advisor and lab IT liaison for review. Their feedback, such as adding a field for "last verified date", was incorporated into the final design, aligning with the structured decision-making process emphasized in the handbook.

Development Phase

The development phase is where the project plan is executed and the system is built. The handbook explains that this stage requires strong coordination and monitoring to ensure alignment with the approved design (Baars, 2006). I assigned tasks based on each team member's strengths. The database developer constructed the database schema, while the UI/UX designer and frontend developer collaborated on the web interface. Meanwhile, the documentation specialist began drafting user guides and technical documentation.

I held weekly progress meetings to ensure alignment with the WBS and to address any issues that arose. Testing occurred in stages: unit testing verified that individual components worked correctly, while integration testing ensured that the interface communicated properly with the database. I also coordinated preliminary data collection by manually inspecting lab computers to compile an initial software inventory.

By following the structured monitoring process outlined in the handbook, I was able to keep the development phase controlled, organized, and aligned with the project's objectives.

Implementation Phase

The implementation phase involves delivering the completed system to users and ensuring successful adoption. The Project Management Handbook describes this stage as the transition from development to operational use (Baars, 2006). I began with a pilot rollout involving two instructors who frequently use the Web Development Lab. They tested the system by verifying software records, adding missing entries, and checking for usability issues.

After incorporating their feedback, such as improving search filters and adding color-coded version alerts, I deployed the system for full use in the lab. I conducted a training session for faculty and the lab IT liaison, demonstrating how to update records and generate reports. I also distributed user guides to support ongoing use. Once the system met all requirements and functioned as intended, I completed the formal acceptance documentation, marking the end of the execution phase.

Follow-Up Phase

The follow-up phase ensures that outcomes are evaluated and lessons are documented. The handbook emphasizes that this phase is essential for closing the project and ensuring long-term success (Baars, 2006). Three months after deployment, I conducted a post-implementation review with faculty and IT staff. The review found that the database significantly reduced confusion about software versions and improved the consistency of lab machines. Students reported fewer issues with mismatched tools, and instructors appreciated having a clear overview of the lab's software environment.

I documented lessons I learned, such as the importance of early access to lab machines and the need for ongoing verification of software versions. I then transitioned long-term responsibility for maintaining the database to the lab IT liaison, ensuring continuity after my team's graduation.

Conclusion

The six-phase model presented in the *Project Management Handbook* (Baars, 2006) provides a structured and practical framework for guiding projects in a higher education institution. By applying the initiation, definition, design, development, implementation, and follow-up phases to the creation of a software-tracking database for the Web Development Lab at Florida Atlantic University, I was able to lead a student team in executing a small-scale but important project.

This structured approach ensured that risks were addressed early, scope remained controlled, and outcomes were evaluated after completion. In a specialized academic environment like the Web Development concentration, such organization is essential for supporting instruction and improving the student learning experience. Grounding the project in Baars' (2006) framework significantly increased the likelihood of both technical and organizational success.

References

Baars, W. (2006). *Project Management Handbook*. DANS : Data Archiving and Networked Services.